Logical transports and the no-push compiler

Supplementary proofs for PA194QNS and TAG196N

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Abstract

This supplement supplies the complete proof translations for exchanging curried premises in three axiom heads and for swapping the quantifier and equality constructor tags. It also gives the matching no-push command compiler with the original sum-based Shift guard. Numerical zero aliases, arbitrary incoming stores, recoding of variable labels and the nonmonotone side condition are treated explicitly. The inherited PA calibration and restricted-implication template are identified as dependencies. These are ordinary mathematical proofs whose syntactic reductions are verifiable in PA, not a new end-to-end Lean or Rocq formalization. The complete physical machines and their executable verification evidence are in the accompanying research archive and main report.

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1 Two exact transports of the numerical proof calculus

This section proves the new logical transformations independently of the literal transition-table construction. A transition table realizes a transformation only after its full emitter and command interfaces have been checked. The proof below is an ordinary mathematical proof that the corresponding syntactic reductions can be verified in PA; it is not a claim that a new end-to-end Lean or Rocq theorem has been compiled.

1.1 The inherited calculus and the precise dependency

Write $p(a,b) = (a+b)(a+b+1)/2 + b$. The standard constructors are

$$I(A,B) = 4p(A,B), \quad M(a,b) = 4p(a,b)+1, \quad U(a,B) = 4p(B,a)+2, \quad E(a,b) = 4p(a,b)+3.$$

Code 0 is truth, code $1 = M(0,0)$ is contradiction, and $N(A) = I(A,1)$. Conjunction and existential quantification are the usual classical abbreviations in I, N, U. The six universally instantiated heads, in their order of emission, are

$$\begin{aligned}
h_1(D,A,H) &= I(\exists x_D\, U(D,H), H),\\
h_2(D,A,H) &= I(U(D,U(A,H)), U(A,U(D,H))),\\
h_3(D,A,H) &= I(M(H,D), I(E(H,A), M(A,D))),\\
h_4(D,A,H) &= I(E(H,D), I(M(A,H), M(A,D))),\\
h_5(D,A,H) &= \exists x_D\, E(H,D),\\
h_6(D,A,H) &= I(U(D,A), I(U(D,I(A,H)), U(D,H))).
\end{aligned}$$

The unused parameter in a head is harmless. In addition the calculus has truth, the dominated implicational schema

$$\rho(D,A,H,d) = I(I(I(H,d),D), I(I(D,H), I(A,H))) \quad (d \le D),$$

the diagonal schema $\Gamma(B) = I(B, U(B,B))$, and the guarded separation gate

$$\begin{aligned}
V(a,b,c) &= I(M(a,b), U(c, M(a,b))), & P(a,b,c) &= I(c, M(c,b)),\\
S(a,b,c) &= U\big(b, (M(a,b) \to P(a,b,c)) \wedge (M(c,b) \to c)\big), & &\\
Q(a,b,c) &= U(c, I(S(a,b,c), N(P(a,b,c)))), & G(a,b,c) &= N(I(V(a,b,c), Q(a,b,c))).
\end{aligned}$$

The gate is an axiom when $c \neq a, b$; a failed guard returns 0. The inference rules are modus ponens and generalization of a theorem $B \mapsto U(i,B)$. These are finite proof trees with explicit axiom parameters and inference annotations.

Call this calculus $\mathcal{C}$. The inherited numerical-calculus calibration provides primitive-recursive proof transformations in both refutation directions between $\mathcal{C}$ and ordinary PA, with correctness provable in PA. Its nontrivial ingredients are the diagonal recovery of reverse membership rewriting and neighborhood equality, the original PA-refutation compiler, and the syntactic PA proof supplier for finite binary comprehension. The source is the primary PA194 report's `calibration.tex` and its attributed reference texts; those arguments are dependencies, not new results of the present emitter change.

The inherited theorem `rho8All` is used at its exact interface: modus ponens, the dominated ρ schema, reflexivity of $\le$, and the bounds $A, B \le I(A,B)$ give a fixed derivation of unrestricted ρ. The same finite template derives the usual K and S schemata, and hence all implicational proof templates used below. It works for any implication operation with the displayed domination bounds, including the code-zero alias $I(0,0) = 0$; no injectivity assumption about implication is required.

1.2 Exchange in the two equality-rewriting heads

Define the exchanged versions

$$h_3^{\mathrm{e}}(D,A,H) = I(E(H,A), I(M(H,D), M(A,D))),$$
$$h_4^{\mathrm{e}}(D,A,H) = I(M(A,H), I(E(H,D), M(A,D))),$$
$$h_6^{\mathrm{e}}(D,A,H) = I(U(D,I(A,H)), I(U(D,A), U(D,H))).$$

Any chosen subset of these three replacements defines a calculus $\mathcal{C}_{\mathrm{e}}$. The third replacement was already used in the supplied Q family; the first two concern the membership-rewriting heads. The emitted diagonal head may likewise be $\Gamma(A)$, $\Gamma(D)$, or $\Gamma(H)$: the parameter is freely selectable in an AX call with zero incoming offset and cache, so every diagonal instance remains available. The same replacement can be combined with an independently checked permutation of a head's three universally quantified parameters: its family of instances is unchanged, and the proof selector uses the inverse parameter permutation.

Explicit proof transformation. The closed combinatory term $\lambda f\, b\, a.\, f\, a\, b$ gives a fixed K, S derivation of

$$I\big(I(A, I(B,C)), I(B, I(A,C))\big). \tag{X}$$

For a completely mechanical proof supplier, bracket abstraction is defined by $[x]x = SKK$, $[x]t = Kt$ if x does not occur in t, and $[x](uv) = S([x]u)([x]v)$. Its usual typing induction is a finite Hilbert proof generator using only K, S and modus ponens. Apply it three times to $f\, a\, b$ with variable types $f : A \to B \to C$, $b : B$, and $a : A$. The delivered implementation produces 61 proof-DAG nodes: 17 K leaves, 14 S leaves, and 30 modus-ponens nodes. Its separate checker verifies every leaf and edge and the closed conclusion. This is a fixed finite derivation of (X), uniform in the numerical formula codes A, B, C. Each of its K, S leaves is replaced by the inherited fixed ρ proof template.

Given an ordinary h_3 axiom, instantiate (X) with $(A,B,C) = (M(H,D), E(H,A), M(A,D))$ and apply modus ponens. This gives h_3^{e}. The analogous instances give h_4^{e} and h_6^{e}. Interchanging A, B in the same template proves the reverse transformations. Every other axiom and every inference is unchanged. Recursion on a finite derivation therefore gives primitive-recursive maps L, R satisfying

$$\mathrm{PA} \vdash \mathrm{Prf}_{\mathcal{C}}(d,B) \Longrightarrow \mathrm{Prf}_{\mathcal{C}_{\mathrm{e}}}(L(d),B),$$

$$\mathrm{PA} \vdash \mathrm{Prf}_{\mathcal{C}_{\mathrm{e}}}(d,B) \Longrightarrow \mathrm{Prf}_{\mathcal{C}}(R(d),B). \tag{E}$$

Proof concatenation, fixed-template instantiation, and traversal of a coded finite tree are primitive recursive. PA verifies (E) by induction on that tree. The transformations preserve conclusions exactly, including code one. Thus they transfer the inherited PA refutation calibration without a new soundness assumption or a change of logical strength.

1.3 An independent residue-permutation construction

The following alternative is a mathematical encoding transformation. It is included even if its literal implementation does not meet the size target; it must not be confused with the accepted table's implemented semantics. Keep I, M unchanged and set

$$U^{\sigma}(a,B) = 4p(B,a) + 3, \qquad E^{\sigma}(a,b) = 4p(a,b) + 2.$$

Replace U, E consistently in every schema and in the emitted generalization. Let σ fix residues $0, 1$ and interchange $2, 3$. Define $F(0) = 0$; for $n > 0$, write $n = 4p(a, b) + r$, $0 \le r < 4$, and put

$$F(n) = 4p(F(a), F(b)) + \sigma(r). \tag{F}$$

For every positive code its two decoded components are strictly smaller. A primitive recursion that stores all earlier values therefore implements (F), so F is primitive recursive. PA proves that $F(n) > 0$ for $n > 0$: if $r \neq 0$ its new residue is nonzero, while if $r = 0$ then $(a, b) \neq (0, 0)$ and induction gives a positive recoded component.

Strong induction, uniqueness of Cantor decoding, and $\sigma^2 = \mathrm{id}$ now prove

$$F(F(n)) = n. \tag{F1}$$

Consequently F is a bijection and preserves equality and disequality of numerical labels. It fixes $0, 1$, and it exchanges $2, 3$. The constructor identities, including the exceptional zero implication, are

$$\begin{aligned} F(I(A, B)) &= I(F(A), F(B)), & F(M(a, b)) &= M(F(a), F(b)), \\ F(U(a, B)) &= U^{\sigma}(F(a), F(B)), & F(E(a, b)) &= E^{\sigma}(F(a), F(b)). \end{aligned} \tag{F2}$$

One must recode numerical variable labels as well as numerical formula codes. Since F is injective, this preserves binding, repeated variable labels, and shadowing. In particular,

$$F(\Gamma(B)) = I(F(B), U^{\sigma}(F(B), F(B))) = \Gamma^{\sigma}(F(B)), \tag{F3}$$

so diagonal self-indexing is preserved exactly. Disequality preservation carries the gate guard to the new guard. Equations (F2)–(F3) carry every other head, modus ponens, and generalization to its corresponding instance.

There is one real obstruction: F is not increasing, since $2 < 3$ but $F(2) = 3 > 2 = F(3)$. Thus an old restricted ρ leaf cannot simply be recoded as a new restricted leaf. Instead instantiate the unrestricted ρ template at the four recoded parameters. Its own restricted leaves have correct numerical side conditions because $A, B \le p(A, B) \le 4p(A, B) = I(A, B)$. The reverse translation uses the same method. Recursion on proof trees gives PA-verifiable maps in both directions, preserving a conclusion B up to $F(B)$. Since $F(1) = 1$, refutations translate both ways. This proves the new numerical calculus is calibrated to PA by composition with the stated inherited calibration.

The same construction can be combined with the head exchanges above. For formula coefficient $k > 4$, use the inherited hereditary radix recoding first and the tag transposition second. On noncanonical residues $4, \ldots, k - 1$, the tag transposition fixes the residue. It is still an involution of all natural codes. The coefficient-change soundness direction must retain the inherited treatment of unused residues; it is not supplied by an inverse map onto the coefficient-four syntax.

1.4 The proof obligation for a literal emitter

The free-constructor checker evaluates the complete emitter word as a stack program, rather than inferring its meaning from the text labels of macros. For the unchanged coefficient-four interface it must prove the pushed list

$$[t,\ U(d, h),\ \rho(D, A, H, d),\ h_1, h_2, h_3, h_4, h_5, h_6, \Gamma(A)]$$

in order of emission, with $D = d + x$, $H = h + y$, and final guarded top $G(H, D, A)$. Exchanged heads or residue-swapped constructors are substituted only where the chosen construction says so. The last

three guard macros must retain their exact effect, including zero output on a failed guard and the cleared temporary registers. This algebraic check is one part of correctness; complete source resolution, both physical decrement outcomes, positive simulation durations, and the entire blank initialization remain separate obligations.

2 A complete command compiler for the changed emitter

This compiler applies to the no-push controller with Shift guard $t+h=1$. It works for any of the exact head exchanges in the preceding section and for either quantifier residue 2 or 3. It does not describe the cached or head-only Shift controllers. The underlying no-push strategy and its reserved-zero discipline are inherited from the supplied PA194N packet; the present application changes which axiom codes the emitter produces.

Let $[a;w]=p(a,w)$ and use nested brackets with the leftmost item at the head. Cantor projections are total and satisfy

$$\mathrm{hd}[a;w]=a,\quad \mathrm{tl}[a;w]=w,\quad [0;0]=0,\quad \mathrm{hd}(0)=\mathrm{tl}(0)=0.$$

These are numerical stacks, not freely generated lists. The command store (t,w,d,h) consists of top formula, stack, offset, and cached formula. With $y=d+x$, the no-push modus-ponens command is

$$\mathsf{MP}_x(t,w,d,h)=\begin{cases}(y,w,y,0), & I(t,y)=h,\\ (0,w,y,0), & I(t,y)\neq h.\end{cases} \tag{MP}$$

Shift has the literal contract

$$\mathsf{S}(t,w,d,h)=\begin{cases}\mathsf{halt}, & t+h=1,\\ (h,\mathrm{tl}(w),d,\mathrm{hd}(w)), & t+h\neq 1.\end{cases} \tag{S}$$

The checked emitter gives

$$\mathsf{AX}_{A,x,y}(t,w,d,h)=(g(H,D,A),[q_1,\dots,q_9,t;w],0,0),\qquad D=d+x,\ H=h+y, \tag{AX}$$

where $q_1=\Gamma(A)$ (or the separately checked $\Gamma(D)$ or $\Gamma(H)$), and $q_2,\dots,q_7$ are the six heads in reverse emission order, $q_8=\rho(D,A,H,d)$, and $q_9=U(d,h)$. A head exchange changes the corresponding q_j and nothing in this stack protocol. All $q_1,\dots,q_8$ and g have implication residue zero; q_9 has quantifier residue 2 or 3 and satisfies $q_9\geq 2$. None is code one.

Direct projection calculation proves

$$\begin{aligned}\mathsf{AX}_{A,x,y};\mathsf{S}^9(t,w,d,h)&=(q_8,[t;w],0,U(d,h)),\\ \mathsf{AX}_{A,x,y};\mathsf{S}^{10}(t,w,d,h)&=(U(d,h),w,0,t).\end{aligned} \tag{A10}$$

Every Shift displayed here is nonhalting: it tests zero against a generated non-one code, two such generated codes, or the universal code of value at least two. The identities hold even when a zero stack cell collapses.

Put $A_0=\mathsf{AX}_{0,0,0}$ and define

$$\mathsf{Push}=A_0;\mathsf{S}^9;\mathsf{MP}_0,\qquad \mathsf{Pop}=A_0;\mathsf{S}^{11},\qquad \mathsf{Discard}=\mathsf{MP}_0,\qquad \mathsf{Offset}_i=\mathsf{MP}_i.$$

Their contracts are

$$\begin{aligned}\mathsf{Push}(t,w,d,h) &= (0,[t;w],0,0),\\ \mathsf{Pop}(t,[a;w],d,h) &= (t,w,0,a),\\ \mathsf{Discard}(t,w,0,h) &= (0,w,0,0),\\ \mathsf{Offset}_i(0,w,0,0) &= (0,w,i,0).\end{aligned} \tag{P}$$

For Push, the final comparison is between an implication code and a universal code, so it fails. Pop's eleventh Shift tests $U(d,h)+t \geq 2$, so Pop is total, including when $t=1$. For Discard, both branches of (MP) coincide at $y=0$. For Offset, $I(0,i)>0$ if $i>0$, while the successful zero case returns the same stated store. Thus no false assumption about $I(0,0)=0$ or $[0;0]=0$ is hidden in (P).

Axiom leaves. Starting from arbitrary $(t,w,0,0)$, first Discard. For $1 \leq j \leq 7$,

$$\mathsf{Discard};\mathsf{AX}_{A,D,H};\mathsf{S}^{j+1};\mathsf{Pop}^{9-j}$$

returns $(q_j,w,0,0)$. The Pops remove exactly the unselected generated tail and the saved zero; the final cache is that saved zero. If an emitted head uses a permutation of its parameters, choose the AX arguments by the inverse permutation for this one selected head. If q_1 is $\Gamma(D)$ or $\Gamma(H)$, set that freely chosen AX parameter to the desired code. No simultaneous choice of parameters for the other unselected heads is needed. For $u \leq a$, the word

$$\mathsf{Discard};\mathsf{Offset}_u;\mathsf{AX}_{b,a-u,c};\mathsf{S}^9;\mathsf{Pop}$$

returns $(\rho(a,b,c,u),w,0,0)$. A permitted gate uses $\mathsf{Discard};\mathsf{AX}_{A,D,H};\mathsf{Pop}^{10}$ and returns $(g(H,D,A),w,0,0)$. Truth uses Discard.

Generalization. From $(a,w,0,0)$, the word

$$\mathsf{Gen}_i = \mathsf{Push};\mathsf{Offset}_i;\mathsf{S};\mathsf{AX}_{i,0,0};\mathsf{S}^{10}$$

returns $(U(i,a),w,0,0)$. Its first three operations produce $(0,w,i,a)$; then (A10) gives the conclusion. Every Shift is safe, including when the premise code is one.

Modus ponens. Suppose recursive words $C(a), C(I(a,b))$ return the displayed conclusion with the stack tail preserved and offset/cache zero, or halt earlier. Use

$$\mathsf{Discard};\mathsf{Push};C(I(a,b));\mathsf{Push};C(a);\mathsf{Push};\mathsf{S};\mathsf{S};\mathsf{MP}_b;\mathsf{Pop}. \tag{C-MP}$$

The initial Push reserves one zero. Immediately before the two Shifts the store is $(0,[a,I(a,b),0;w],0,0)$. The first Shift exposes a in the cache; the second halts if $a=1$, otherwise giving $(a,[0;w],0,I(a,b))$. Successful MP gives $(b,[0;w],b,0)$ and total Pop gives $(b,w,0,0)$. These numerical projection equations also cover $a=b=w=0$. An early halt already fulfills the refutation-search goal.

Induction on a finite proof tree now proves that $C(p)$, from every $(t,w,0,0)$, either halts or returns $(\mathrm{concl}(p),w,0,0)$. The proof word is primitive recursively constructed. From arbitrary incoming d,h, prepend A_0 to clear both. If p concludes code one, append Shift. Thus

$$A_0;C(p);\mathsf{S} \tag{W}$$

halts from every incoming numerical store. The derivation induction and all displayed calculations are formalizable in PA.

The carried-store scheduler. The scheduler need not reset the store between candidates. For a positive continuation encoding $E(\epsilon) = 0$ and $E(c :: w) = 1 + \text{encodeCommand}(c, E(w))$, its AX code is

$$\text{encodeCommand}(\mathsf{AX}_{A,x,y}, z) = p(A + 2, p(y, p(x, z))).$$

Let $z = E(W)$ for the universal halting word (W). Prefixing it by $\mathsf{AX}_{k,0,0}$ gives fresh candidate codes

$$p(k + 2, p(0, p(0, z))).$$

They are strictly increasing and unbounded in k, and all such prefixes are harmless because (W) works from every store. Hence the scheduler, whose fresh counter begins at two and increases through all later natural numbers, eventually executes a halting word. Positive continuations strictly decrease and the source arithmetic macros terminate. Candidate coding, source-trace formation, and the finite arithmetic bounds used here are primitive recursive; PA verifies the construction by induction.

Soundness and the exact conclusion. From the clean seed, annotate t, h and every live canonical stack entry with a proof in the changed calculus. The zero stack supplies truth, and a zero push on zero adds no canonical cell. AX adds its checked axiom heads, a generalization of the cached theorem, and the saved top. MP applies its two premise annotations or returns truth. Shift transports annotations; if it halts, $t + h = 1$ says one of its two annotated codes is one. Therefore a source halt yields a changed-calculus refutation. The two calculus transports above turn such a refutation into an inherited calculus refutation and then a PA refutation. Conversely, a PA refutation passes through the inherited calibration, the new transport, the compiler (W), and the scheduler to a source halt.

If the literal machine M has the separately verified complete startup and universal positive-duration source simulation, concatenate finite segments in the forward direction. In the reverse direction, a physical halt at time T cannot precede $T + 1$ nonhalting source instructions, since each consumes positive physical time and their simulations contain no premature halt. Determinism gives the required contradiction. Thus PA proves the two finite syntactic reductions and consequently

$$\text{PA} \vdash \text{Halt}(M, \text{blank}) \longleftrightarrow \exists p \, \text{Prf}_{\text{PA}}(p, \ulcorner 0 = 1 \urcorner),$$

$$\text{PA} \vdash \neg \, \text{Halt}(M, \text{blank}) \longleftrightarrow \text{Con}(\text{PA}).$$

This conclusion uses the explicitly identified inherited calibration and its ordinary mathematical proof; it does not upgrade an executable Python certificate into a proof-assistant theorem, and it does not assume PA consistency in order to prove the biconditional.